MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSCNBC200– NUTRITIONAL BIOCHEMISTY

Duration: 2 hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Each question is tagged with the appropriate Bloom's Revised Taxonomy level (Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating.) and the relevant course outcome (CO) number.

* Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S).

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries **1** mark

1. Name the enzyme involved in the conversion of glucose to glucose-6-phosphate [U] [1]

   a) Kinase b)Isomerase c)Mutase d) Enolase

2. Cori cycle is also called…….[U][1]

   a) TCA cycle b) Lactic Acid Cycle c) Tricarbocylic Acid d) Pentose Phosphate Pathway

3. Name the protein that accepts glucose from UDP glucose. [U] [2]

   a) Glycogen synthase b) UDP Phosphorylase c) Glycogenin d) GALT

4. Name the cycle involved in eliminating toxic ammonia from the body.[K] [2]

   a) Transamination b) Deamination c) Urea Cycle d) Transdeamination

5. Name the organelles where β-oxidation of fatty acids takes place.[K] [3]

   a) Cytoplasm b) mitochondrion c) Nucleus d) Endoplasmic Reticulum

6. Name the enzyme is responsible for triglyceride breakdown in adipose tissue [K] [3]

   a) Lipases b) Thiokinases c) dehydrogenase d) Thiolase

[1 X 6 = 6 marks]

Part B

Short Answer Type Questions

Answer **2** Questions

Each question carries **2** marks

7. Explain Cori Cycle [U] [1]
8. Describe Fructolysis. [U] [3]
9. Explain the term ketogensis.[U] [3]
10 Classify the different types of adipose tissue? [K] [3]

[2 X 2 = 4 marks]

Part C

Short Essay Type Questions

Answer **4** Questions

Each question carries **5** marks

11. Explain Citric acid cycle. [U] [1]
12. Discuss the energetics of breakdown of a single glucose molecule. [U] [1]
13. Describe urea cycle. [U] [2]
14. Differentiate between Transmination and Deamination [U] [2]
15. Explain synthesis of triacylglycerol [U] [3]
16. Differentiate between Betaoxidation and Synthesis of Fatty acid [R] [3]

[5 X 4 =20 marks]

Part D

Essay Type Questions

Answer **2** Questions

Each question carries **10** marks

17. Explain Gluconeogenesis [U] [1]
18. Describe the various reactions of amino acid metabolism.[U] [2]
19. Explain the metabolism of fattyacid[U] [3]
20. Explain the metabolism of Cholesterol[U] [3]

[10 X 2 =20 marks]